

Transforming Emissions Into Sustainable Materials

A Modular Closed-Loop Carbon Management Platform for the \$10 Trillion Construction Sector.

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The Dual Challenge: Maintaining profitability while navigating strict new environmental regulations.



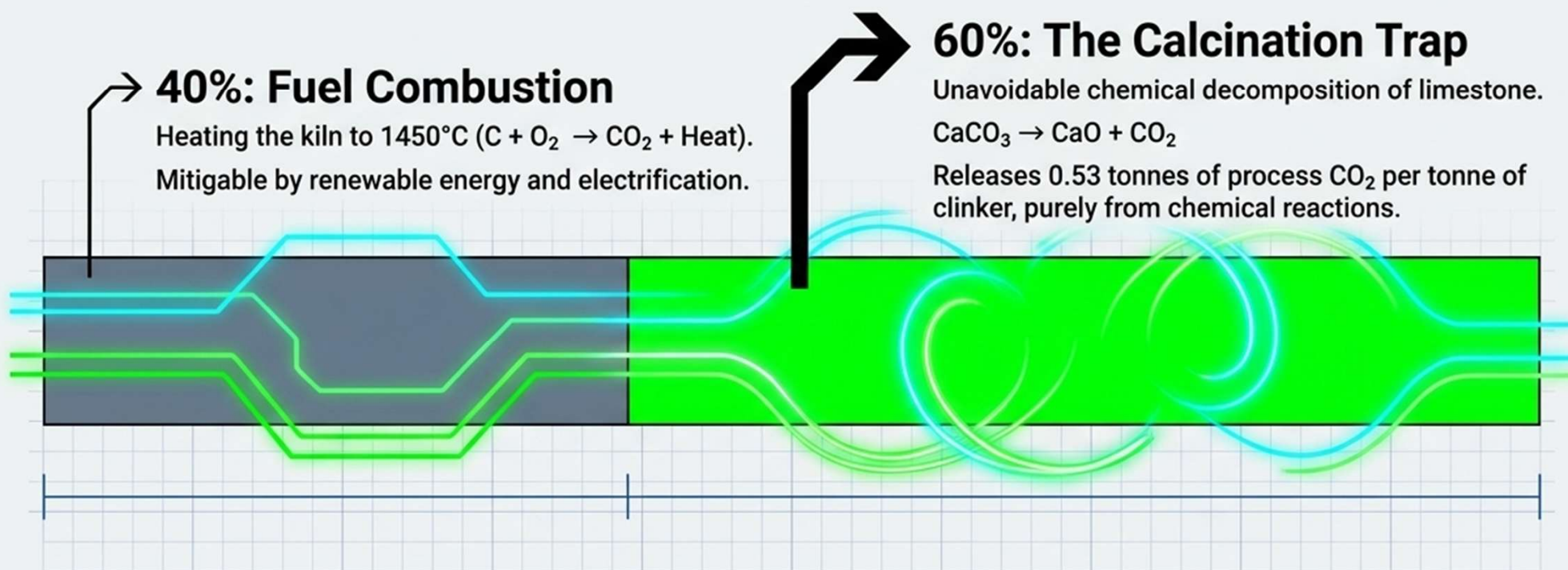
The Material Gap: Existing building materials and methods are insufficient to meet global net-zero goals.



The Market Pressure: Surging demand from stakeholders for eco-friendly, ESG-compliant solutions.

40% of global **CO₂ emissions** originate from the construction sector.
(with **8%** from concrete alone)

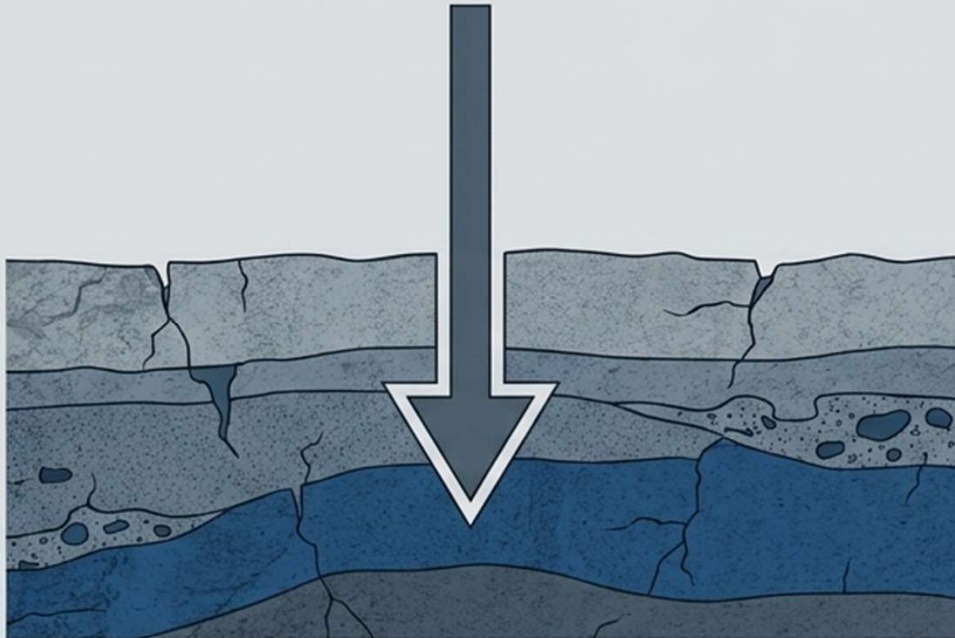
The Hidden Bottleneck: Why Solar Panels Cannot Fix Cement



Synthesis: 60% of cement emissions are chemical, not fuel-based.
Active Carbon Capture is a physical imperative.

The Old Paradigm

Carbon Capture & Storage (CCS)



- Waste Disposal Model
- Represents massive sunk operational costs.
- Requires heavy transport infrastructure (pipelines).
- Demands permanent regulatory monitoring.

The New Paradigm

Carbon Capture & Utilization (CCU)



- Value Creation Model
- Transforms an industrial liability into a premium manufacturing input.
- Enables 100% on-site utilization.
- Manufactures high-margin, carbon-negative concrete and bioplastics.

The Modular Closed-Loop CCU Platform

1. Intercept

Capture CO₂ directly from point-source exhaust (flue gas) before atmospheric release.

2. Purify & Compress

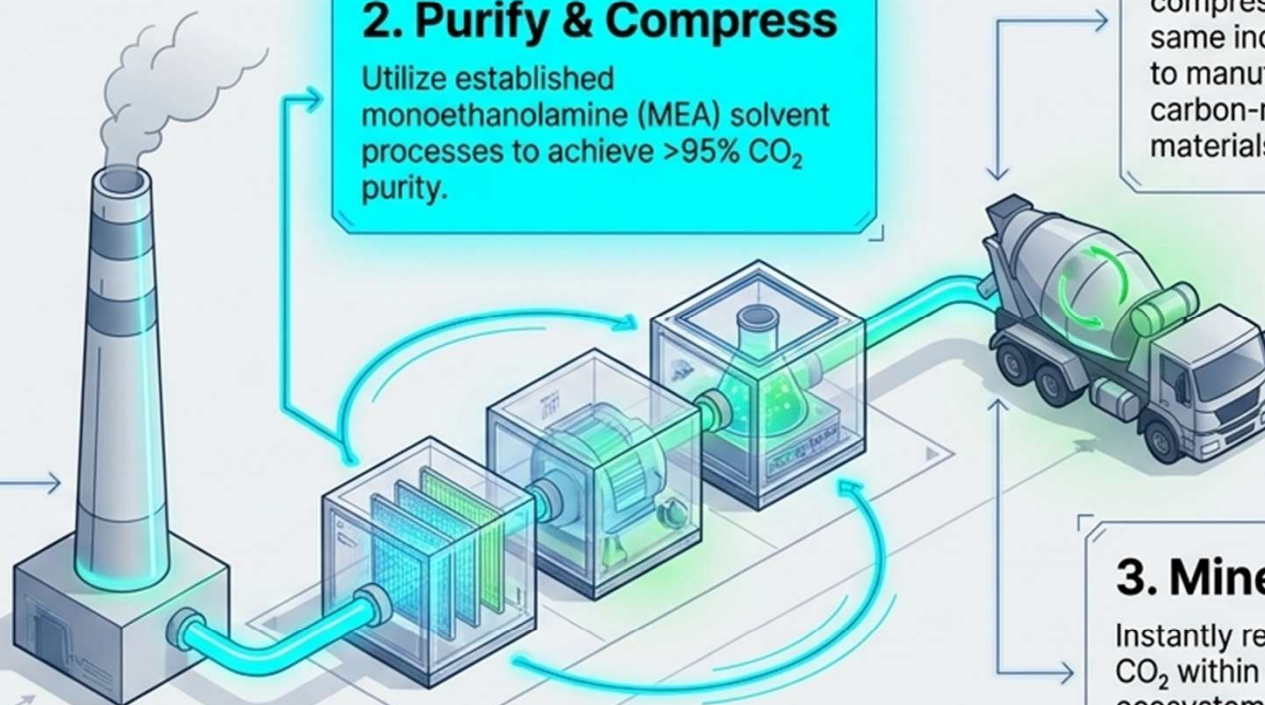
Utilize established monoethanolamine (MEA) solvent processes to achieve >95% CO₂ purity.

3. Mineralize

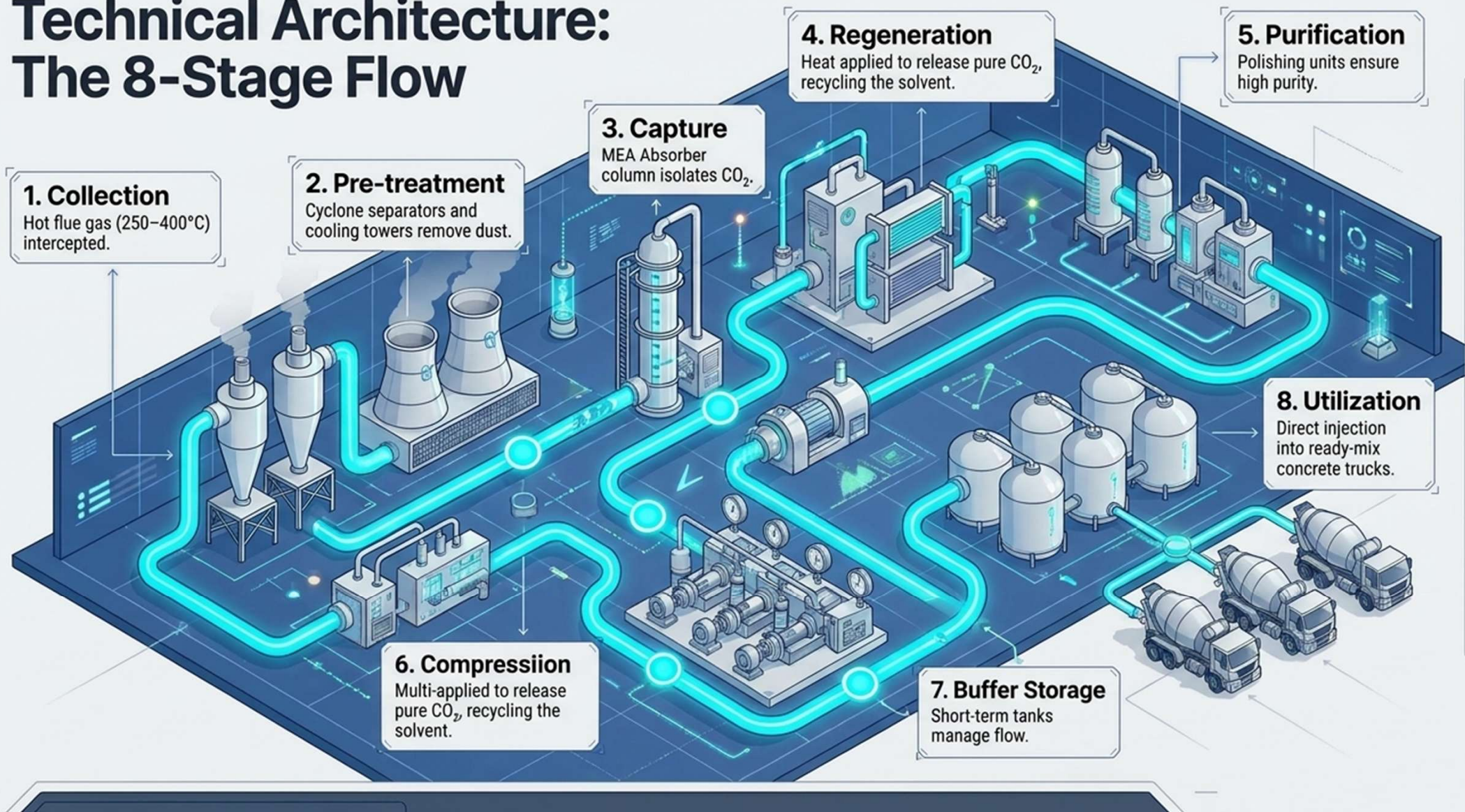
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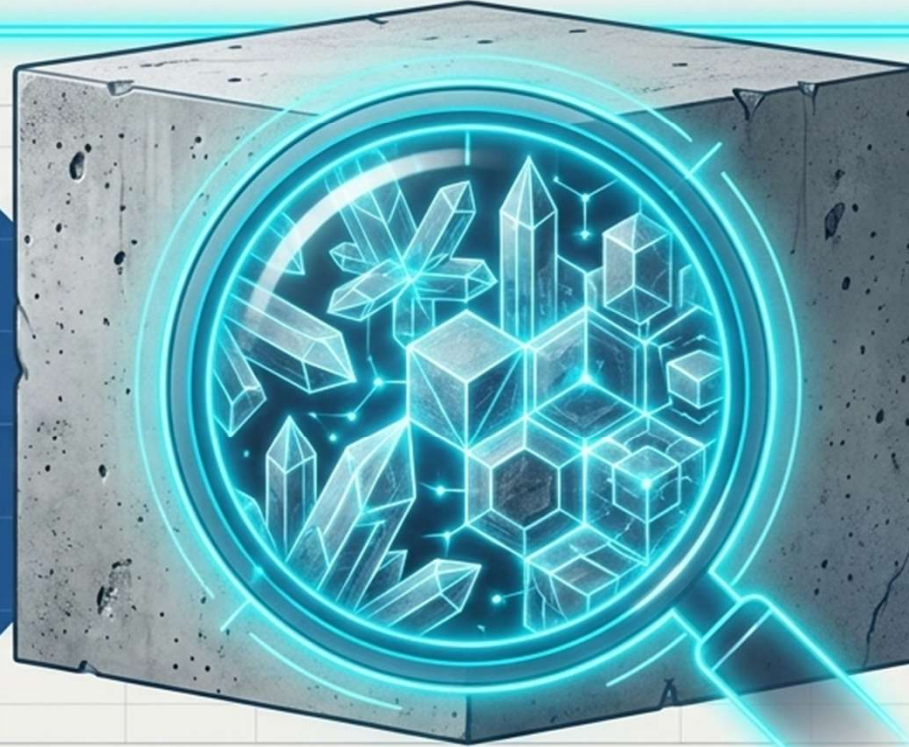
Technical Architecture: The 8-Stage Flow



The Chemical Miracle: Mineralization

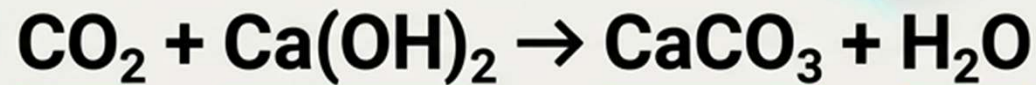
Permanent Locking

Carbon is chemically transformed into solid rock. Re-release into the atmosphere is structurally impossible.



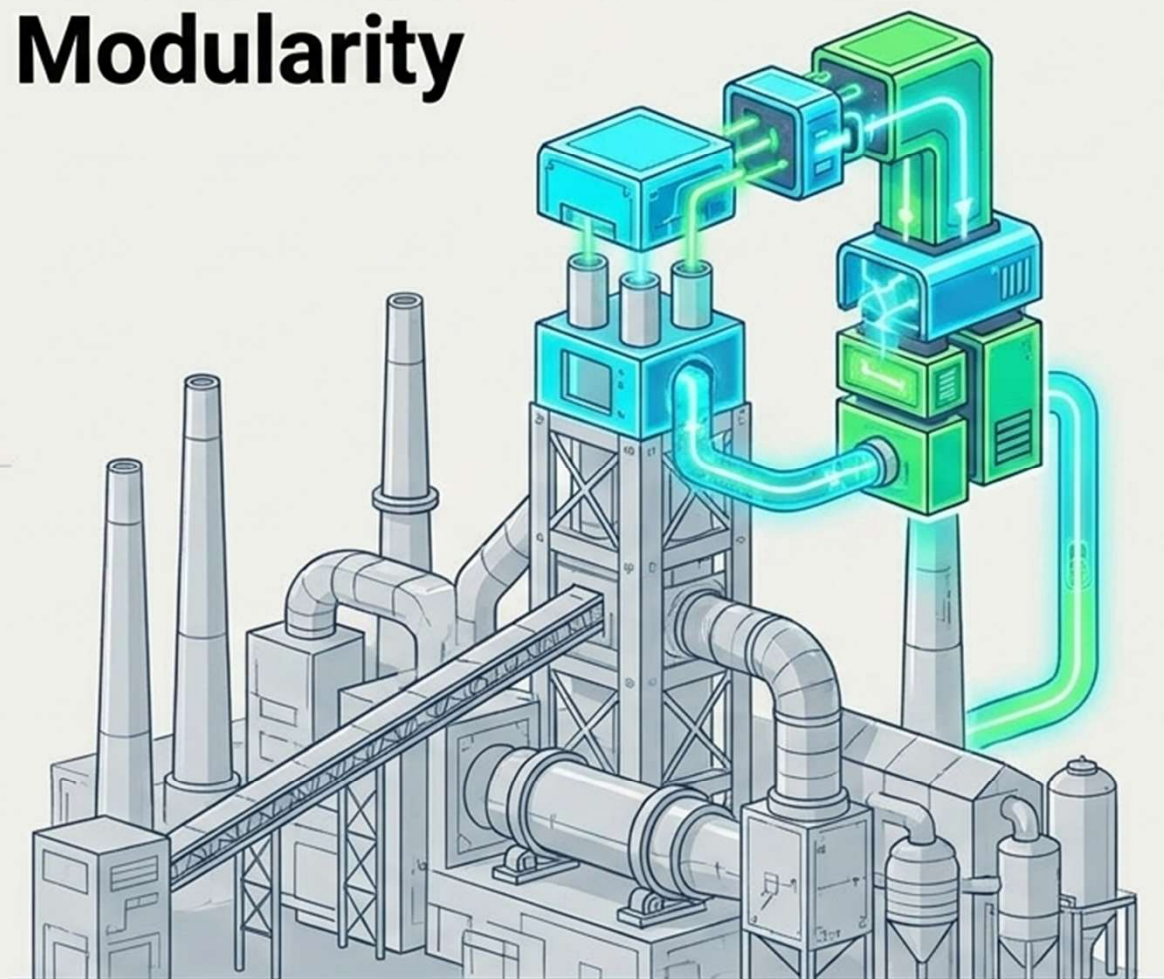
Structural Superiority

The resulting CaCO_3 microscopic crystals increase concrete density, drastically reduce porosity, and enhance long-term compressive strength.



We are not just hiding carbon; we are engineering a superior construction material.

The Innovation: Modularity



CapEx Efficiency

Zero need for total plant redesign or expensive structural teardowns.

Zero Downtime

Retrofit compatibility means physical installation happens without halting clinker production.

Infinite Scalability

Start with a single capture unit; snap on additional modules seamlessly as production volumes scale.

Easier Maintenance

Isolated system modules allow for targeted upgrades and repairs without systemic plant shutdowns.

Competitive Diagnostic

Dimension	Conventional CCS	Our Modular CCU
End State	Underground disposal	High-value product (CaCO_3)
Revenue Impact	Massive sunk operational cost	Drives new product revenue streams
Infrastructure	Requires expensive pipelines	100% On-site utilization
Deployment	Massive bespoke engineering	Plug-and-play modular retrofits
Carbon Status	Requires indefinite monitoring	Permanently mineralized into rock

The Market Opportunity

TAM: >\$10 Trillion

Global Construction Market



CAGR: 8.5%

Sustainable Materials Sector
Growth (Next Decade)

Market Drivers



– Stricter global environmental regulations and carbon taxation.



– Surging global urbanization demanding massive new infrastructure.



– Corporate ESG compliance mandates forcing supply chain decarbonization.

The Business Model: Driving B2B Profitability

Revenue Engine

One-Off Revenue (CapEx)

-  **Equipment Sales:** High-margin modular CCU units.
-  **Installation Services:** Engineering, site prep, and commissioning.

Recurring Revenue (OpEx & SaaS)

-  **Maintenance Contracts:** Annual agreements for solvent management and replacement parts.
-  **Software & Monitoring:** SaaS dashboard tracking live CO₂ captured/utilized, energy consumption, and systemic efficiency.
-  **Carbon Accounting:** High-value consulting and verified ESG reports for regulatory compliance.


High LTV Streams

The B2B Value Proposition



Environmental Value

Delivers an immediate reduction in industrial footprint and creates a permanent carbon sink, directly satisfying corporate sustainability and net-zero goals.



Economic Value

Avoids harsh carbon compliance penalties and taxes, allows the sale of premium-priced low-carbon concrete, and drastically cuts transport costs via 100% on-site utilization.



Operational Value

Non-disruptive retrofit design ensures continuous clinker production with seamless modular expansion capabilities.

Implementation Roadmap



Phase 1: R&D & Design

Finalize systemic design, run operational process simulations, and conduct strict lab validation on capture technology.



Phase 2: Pilot Prototype

Install small-scale demonstration unit, validate capture efficiency under real-world conditions, and test concrete quality.



Phase 3: Industrial Pilot

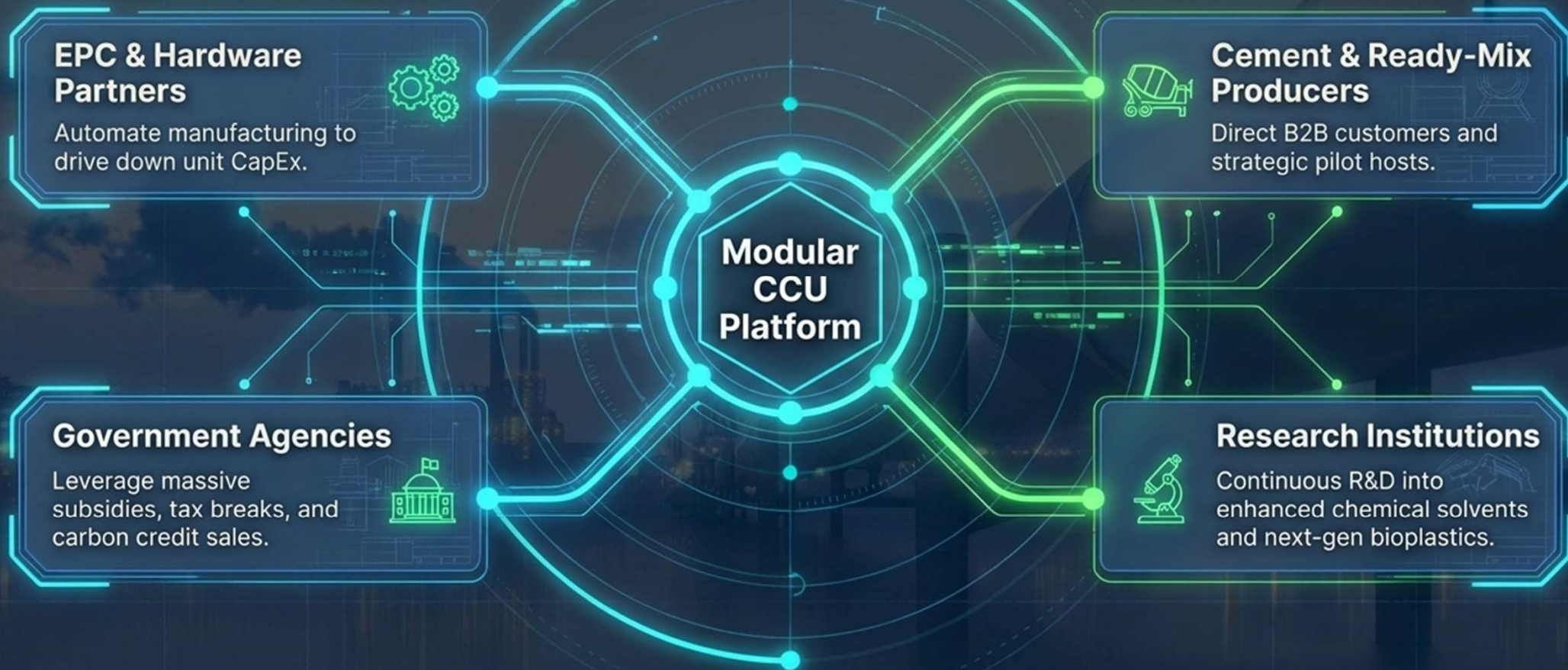
Deploy at a strategic partner cement facility. Collect live operational data and assess real-world economic unit economics.



Phase 4: Commercial Deployment

Launch modular installations to B2B customers globally, roll out SaaS monitoring platform, and expand maintenance operations.

Scaling & Ecosystem



The Imperative: Climate Action

Turning **40%** of global emissions from a planetary liability into a closed-loop, carbon-neutral certainty.



The Opportunity: Economic Return

A **scalable, modular hardware + SaaS** business model entering a **\$10 Trillion** market undergoing a forced regulatory transition.



Investing in the Modular CCU Platform is a singular opportunity to drive the construction industry's net-zero transition while securing unprecedented B2B recurring revenue.